Linear Regression Analysis of New York City TLC Data

Summary Report – Project 5
Automatidata – NYC TLC 2017

Project Overview

This phase of the project focuses on constructing, evaluating, and interpreting a multiple linear regression model to predict NYC taxi fares based on existing data variables.

Key Insights

The Problem: The goal of this project phase was to build a multiple linear regression model to predict taxi fares using existing NYC TLC data.

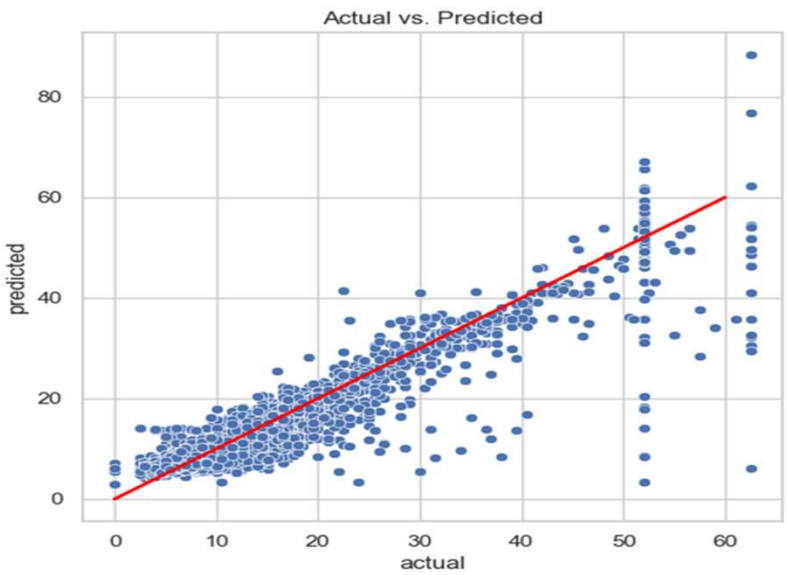
Proposed solution: By analyzing relationships between a continuous dependent variable (fare amount) and several independent predictors, the model achieved a high level of predictive accuracy.

Keys to success

- New variables were created, such as mean_distance and mean_duration, to capture geographic trends based on unique pickup and drop-off location pairs.
- All features were scaled using StandardScaler to ensure that variables with different units were treated with equal mathematical weight during model training.

Details

The visualization displays a strong alignment between the model's predictions and the actual fare amounts, with data points clustering tightly around the x=y line of best fit.



Alt Text: New York City TLC data scatter plot graph showing actual vs predicted fare amounts.

Next Steps

- Future versions of the model can skip predictions for known flat-rate trips, such as the $52 JFK airport rate, by directly imputing those values based on the `RatecodeID`.
- Transition from linear regression to tree-based machine learning models (like Random Forest) to capture non-linear relationships and potentially improve predictive performance further.